

Multimodal Emotion Recognition in Social Robotics: Benchmarking Edge LLMs for Low-Latency Human-Robot Interaction



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MSci (Hons)

A dissertation submitted for the degree of
Master of Science in Computer Science

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June, 2026

Declaration

I declare that the work presented in this dissertation is, to the best of my knowledge and belief, original and my own work. The material has not been submitted, either in whole or in part, for a degree at this, or any other university. Estimated word count is: **11858**

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Abstract

Effective Human Robot Interaction (HRI) in social robotics requires systems to dynamically adapt to human’s emotional state’s. While Large Language Models (LLMs) offer advanced emotion reasoning capabilities, deploying them in embedded systems introduces two critical engineering challenges: computational latency and ”modality laziness”. This dissertation addresses these limitations by designing, implementing and benchmarking a lightweight, low latency multimodal system that utilised a late-fusion pipeline whilst running an LLM natively on the LuxAI QTrobot. The system decouples audio and video streams using ROS (Robot Operating System), processes the real-time speech into transcription via Nvidia Riva, in parallel with facial expression extraction using DeepFace. These visual metrics are compressed into low dimensional data metrics and then stored in a rolling buffer. The buffer is fused with text transcripts at the point of decision, entirely passing the raw visual token overhead.

Seven open-source edge LLMs were benchmarked under varying prompt constraints using a subset of the MELD dataset. The evaluation exposed a severe ”Multimodal Penalty”, where models struggled to reconcile accurate text with noisy visual data. Macro F1 scoring revealed that larger models, such as Mistral 7B, masked this representational failure through the majority class illusion. Mistral 7B classified neutral predominantly while smaller models collapsed into negative sentiment loops hyper-fixating on ”sadness”. Despite these reasoning limitations Qwen2.5 3B emerged as the optimal choice for decision making in the QTrobot. It successfully balanced a sub-second processing latency of 0.79 seconds with 100% formatting rate, ensuring stable, real-time control of the physical robot gestures and facial expressions. This research demonstrates a multimodal architectural framework that is responsive on resource constrained, offline systems.

All code can be viewed here: <https://github.com/jackhodgkins/EmotionalRobot.git>

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Chapter 1

Introduction

1.1 Study Background

In the field of social robotics, effective Human-Robot Interaction (HRI) requires systems to move beyond the rigid execution of task-oriented commands and sequential programming. For companion, healthcare, and general social aspects, the ability for the system to recognise and dynamically adapt to the user’s emotional states is a primary feature for long term engagement and overall interaction quality.

Historically, affective robotics relied heavily on unimodal frameworks. However, natural human communication is multimodal; emotional intent is expressed through a simultaneous combination of verbal semantics and non-verbal cues. More recent advancements in Large Language models (LLMs) have opened new frontiers for social robotics, shifting the paradigm from basic rule based classifiers to more sophisticated ”emotional reasoning”. When integrated into an interactive loop, locally deployable models enable robotic platforms to act as empathetic conversational agents, capable of evaluating complex nuances, and occasions conflicting user states in real time.

1.2 Problem Statement

Deploying conversational, socially intelligent robotic system introduces two critical, conflicting engineering challenges: conversational latency and cross-modal attention bias.

1. **The Latency Threshold:** In natural human conversation, response delays that would often exceed 2 seconds disrupt the natural flow of conversation and shatter the immersion for the user. Processing high-dimensional, raw visual data alongside real-time automatic speech recognition locally on embedded hardware exceeds traditional processing limits, this results in severe conversational bottlenecks.

2. **Modality Laziness (Text Bias):** Large Language Models are predominantly pre-

trained on massive textual datasets. This makes them highly capable at processing linguistics. When deployed in a multimodal fusion architecture, these models have been known to frequently display "modality laziness", defaulting to the path of least resistance by basing decisions entirely of textual transcripts and ignoring structure numerical inputs such as facial metrics.

1.3 Research Aims and Objectives

This dissertation addresses these constraints by designing, implementing and systematically benchmarking a low-latency multimodal late fusion pipeline running locally on LuxAI QTrobot. The core architecture focuses on bypassing raw visual data and instead converting the video feed and analysis into low-dimensional numerical facial metrics that are store in a rolling ring buffer and then fused with real-time audio transcriptions at the point of decision.

To validate this approach, the research aims to fulfill the following objectives:

- **Pipeline Orchestration:** Developing a parallel processing architecture using Robot Operation System (ROS) middleware to handle concurrent audio transcription (via Nvidia Riva) and facial expression analysis (via DeepFace) without blocking each other.
- **Prompt-Engineered Mitigation:** Design a dual-phase prompting framework to evaluate explicit instructions impact the LLMs ability to identify conflicts (cross-modal mismatches) and adhere to strict hardware commands for controlling the system.
- **Edge LLM Benchmarking:** Execute and in depth benchmarking study on seven open-source language models in order to understand and isolate the occurrence of the "Multimodal Penalty" using strict classification accuracy and then Macro F1 scoring.
- **Operational Validation:** Analyse the trade-off between model parameter size, format compliance, and response latency to determine the most suitable model for real-time HRI deployment within the QTrobot.

1.4 Dissertation Structure

The reminder of this report will be structured as follows:

- **Chapter 2 (Literature Review):** Exploring the foundational theories researching expression coding, text base sentiment analysis, and multimodal fusion paradigms.
- **Chapter 3 (System Design):** Outlining the conceptual framework of Sense-Think-Act. Detailing node topology, rolling buffer mechanics and inline markup conventions for hardware commands.

- **Chapter 4 (Implementation):** Presenting the technical details of the software stack, breaking down how Nvidia Riva, DeepFace, and the custom Python middle ware that handles parsing and running the models locally via Ollama.
- **Chapter 5 (System in Operation):** Describing the runtime state machine and execution sequence of the live Human-Robot Interaction.
- **Chapter 6 (Testing and Evaluation):** Details the empirical findings from the MELD dataset (Poria et al., 2019), exposing majority-class illusion, smaller model’s behavioural collapse loops, and operational latency across the benchmarked models.
- **Chapter 7 (Conclusion and Future Work):** Summarising the primary contributions of this research and the dissertation. Outlining the clear pathways for future research.

Chapter 2

Background

2.1 The Role of Emotion in Human-Robot Interaction

Effective human-robotic interaction extends between just beyond the execution of tasks. In the world of social robotics it is essential for these systems to be able to respond to their user's emotional states accordingly. This is particularly critical in the context of healthcare, therapy, and companionship. In these particular scenarios, the ability of the robot to be able to respond empathetically to a user's emotional state directly influences user engagement, trust and overall quality of the interaction.

Emotion is inherently a multi-modal occurrence. Humans use both the context and sentimental meaning of the words, as well as the visual facial expressions that the person expresses to understand emotional state. Due to this, for any system in which the aim is to reproduce the multimodal process, attempting to replicate the process in a systematic way needs to be considered for these artificial systems. For example, if a user says. "I am fine." but proceeds to be frowning, the system needs to understand that in fact the verbal cues are misleading and that alone the system may then misclassify the emotional state of the user.

2.2 Facial Emotion Recognition: Approaches and Limitations

The human face is one of most informative queues when it comes to non-verbal communication. A person's face is able convey emotions such as social intent and cognitive state in ways that are essential for communication between people (P. Ekman and Friesen, 1978). Facial Expression Recognition (FER) is the systematic task of identifying and classifying these signals from images and or videos. Research in this domain has progressed considerably over the last three decades. Understanding and being able to apply some of these approaches is crucial to this project.

2.3 The Facial Action Coding System (FACS)

Early work in automated FER was grounded in the Facial Action Coding System (FACS). This framework developed by P. Ekman and Friesen (1978) breaks down the facial movements into Action Units (AUs). These are movements within the face that are detected and assigned this tag, for example if the inner brow is raised it may assign the tag AU1, the brow being lowered = AU4. Lien et al. (1998) created a FACS-based computer vision system capable of extracting more macro expressions from the human face. This was done via three complementary extraction methods: facial feature point tracking, dense optical flow combined with Principal Component Analysis (PCA), and high gradient component analysis in the spatio-temporal domain. The classification was then performed using Hidden Markov Models (HMMs) to model the temporal dynamics of expression sequences. This began to achieve rates of up to 93% for upper-face expressions on controlled lab data. Importantly, this work proved that most expressions encountered in everyday life couldn't all be classified as the six prototypical emotions (joy, anger, fear, disgust, sadness, and surprise). Instead, a more subtle combination of these AUs vary in intensity and duration. This distinction between posed and spontaneous expressions remains one of the central challenges to FER research.

2.4 Overcoming Pose Variation and Occlusion on Unconstrained Environments

A noticeable limitation of these classical FER systems, including those trained on the earlier datasets, is their reliance on controlled conditions such as consistent lighting, frontal faces, exaggerated expressions etc. These conditions can be near possible to control in a real-world environment. Head position variation often presents a major obstacle as the user may rotate their head position away from the frontal plane. This can cause partial self-occlusion of their facial features and dramatically alter the appearance of AUs. K. Wang et al. (2020) directly addressed this issue by proposing a Region Attention Network (RAN) that learns to adaptively weight different facial regions based on which are visible and un-occluded, demonstrating that pose-robust and occlusion-robust FER requires explicit architectural considerations. This is especially relevant to Human-Robotic Interaction where neither the user's head pose, nor the robot's camera angle can always be fully controlled.

2.5 Deep Learning for Micro-Expression

The rise of deep learning models brought substantial performance gains to FER, particularly for macro-expressions (0.5 seconds to 4 seconds). However, the recognition for micro-expressions (less than 0.5 seconds) remains to be a considerably harder problem. Nguyen et al.

(2023) introduced Micron-BERT, a self-supervised transformer-based architecture specifically designed for micro-expression recognition. The model utilises two key mechanisms: Diagonal Micro-Attention (DMA) to detect the subtle inter-frame differences, and Patch of interest (PoI) to localize and highlight facial regions whilst also suppressing background noise. This was then trained on up to eight million unlabeled images and evaluated on four benchmarks including CASEME II and SAMM. Micron-BERT achieved state of the art recognition performance, outperforming prior methods by substantial margins. While Micron-BERT demonstrates what is achievable when a model is deigned specifically for micro-expression recognition, its computational demands and requirements for high-speed video input (upwards of 200 frames per second) makes it difficult for implementation into the QT Robot’s camera with a maximum framerate of 90 frames per second.

2.6 OpenFace

For a system acting in a live human-robot interaction, real-time facial expression analysis is required without the need of specialised hardware. OpenFace 2.0 (Baltrusaitis et al., 2018) represents a significant practical contribution. OpenFace 2.0 is an open-source toolkit that performs facial landmark recognition with in a unified real-time pipeline. The system uses a Convolutional Experts Contained Local Model (CE-CLM) for landmark detection, which is then explicitly trained across multiple head orientations and is scaled to handle non-frontal faces and self-occlusion, this is a key advantage over systems that assume near frontal input. Critically, OpenFace 2.0 estimates head poses alongside the AU detection. This allows it to account for geometric distortions that are introduced by head movement rather than just treating it as a source of error. OpenFace 2.0 achieves landmark detection at 30-40Hz on standard CPU hardware without the need for GPU acceleration. This satisfies the real-time requirement of interactive robotic systems. OpenFace’s training on spontaneous datasets rather than posed ones is also a large distinction, unlike tools like DeepFace, whose emotion classifier was trained the posed expression databases. OpenFace 2.0’s AU models are calibrated to the subtler, less exaggerated expressions encountered in natural human interaction.

2.7 DeepFace

DeepFace (Serengil and Ozpinar, 2021) is an open-source Python library that wraps several deep learning models for facial recognition and emotion classification, including VGG-Face, FaceNet, and a dedicated emotion recognition model. It provides a high level API that is capable of returning discrete emotion predictions. These are happiness, sadness, anger, fear, disgust, surprise, and neutral. This is all done from a single image frame with minimal configuration required.

2.8 Linguistic and Semantic Emotion Analysis

As important as facial expressions are in understanding someone’s emotional state, the semantic and linguistic content of their speech is equally as important. To maximise the capabilities of the QT Robot we need to develop the system in such a way that it is able to interpret this. The following section delves into text based sentiment and emotion analysis in order to assist in the development of the system that will be able to interpret human affect.

2.9 What is Sentiment Analysis and Why Does it Matter?

Sentiment analysis is the computational task of identifying and extracting the subjective information from text. This is most commonly to understand the polarity of opinion in the text (positive, negative, or neutral) (Pang and Lee, 2008) . Liu (2020) offers a more broad framing, describing it as the analysis of people’s opinions, sentiments, evaluations, appraisals, attitude, and emotions expressed through language. In a social robotic system it essential for the robot to be able to understand the semantics behind the user’s speech rather than just treating all inputs as emotionally neutral.

For social robotics specifically, the ability to interpret and understand what the user is saying is fundamental to the system’s ability to then be able to respond to the user in an empathetic manner. A robot that responds to the user without regard for its own tone or expressions risks appearing cold.

2.10 Transformer based models and LLMs

The introduction of transformed based models, particularly BERT (Devlin et al., 2019) and its successor, significantly advanced the field. Pre-training on large unlabelled data sets and fine tuning on labelled affective datasets means that these models capture long-range semantic dependencies that rule-base systems previously struggled with. As of more recently, large language models (LLMs) such as LLama and Gemma have demonstrated strong emotion classification capabilities through zero-shot and instruction based prompting. This means LLMs don’t require specific task specific tuning (Chae and Davidson, 2025). Kadiyala (2024) showed that 4-bit quantised versions of Gemma-7B and LLaMa-3-8B achieved state of the art results at the WASSA 2024 emotion detection task, this demonstrates that these locally deployable models are now competitive for affective emotional classification.

2.11 Emotion Classification from Sentiment Analysis

It is important to distinguish the difference between sentiment analysis and emotion classification. Sentiment analysis often reduces the text's affect to a positive/negative/neutral polarity, where as emotion classification targets discrete affective categories such as Paul Ekman (1992) six basic emotions: anger, disgust, fear, happiness, sadness and surprise. Thus for a social robot that must adapt its tone and behaviour based on the user's emotional state, this finer granularity is necessary; knowing that a user is expressing negative sentiment is insufficient if the appropriate response to anger and sadness are different.

2.12 Limitations

Text based emotion classification faces several limitations regardless of model. Sarcasm and irony are particularly problematic as the initial word choice often contradicts the actual intended affective meaning of the words (Joshi, Bhattacharyya, and Carman, 2016). Context dependence is a further challenge as the same utterance can often carry different emotional weights depending on the speaker and conversational history. This is why turn based classifiers can often struggle. On top of this, transcription issues in the Speech to Text (STT) can often throw classifiers off as words with specific sentimental meaning may be picked up even if the user did not say it. These limitations motivate the multimodal approach taken in this project where the linguistic analysis combined with the facial expressions recognition will make the classification rather than just the stand-alone inputs.

2.13 Multi-Modal Emotion Recognition

2.14 Why Multimodal Computing?

When creating an affective HRI system that needs to have social capabilities then having on unimodal channel can introduce significant blind spots for the system. The Unimodal frameworks can often cause performance issues, especially when in unconstrained environments. Facial Expression Recognition (FER) alone can struggle with subtle macro-expressions or head pose variations, where as on the other hand text-based models are entirely blind to the linguistical tone and vocal tone as well as any visual characteristics. Conversely, Audio Speech Emotion Recognition (SER) models can frequently misinterpret raw signal pitch changes when stripped of the language context. The idea of using a Multimodal fusion approach addresses these systematic failures by integrating complementary input streams slightly matching the cognitive workflows that humans naturally utilise to decode the emotional intent of a person.

2.15 Fusion Strategies

The integration of these distinct, complementary modalities is often categorised into three foundational paradigms (Baltrušaitis, Ahuja, and Morency, 2018):

- **Early Fusion (Feature-Level):** Raw features pulled from low-level vectors. The modality is concatenated directly into a singular, unified input later prior to classification. This preserves the features in a highly granular way, however it can introduce massive input dimensionality issues as it tries to synchronise different sampling rates such as matching high amounts of visual frames with a significantly smaller amount of textual input.
- **Late Fusion (Decision-Level):** Running completely separate processes at once allowing for the data collection of voice, facial expressions, text to all be independent of each other, then each process making its own decisions. However, this completely ignores how the modalities each affect each other midway through the interaction.
- **Hybrid/Intermediate Fusion (Representation-Level):** In the hybrid approach we let the different processes make their decisions, collect the data and then combine the outputs together to then finally classify the decision.

A major risk in modern Hybrid Fusion approaches is something known as "modality laziness" (W. Wang, Tran, and Feiszli, 2020). Due to Large Language Models (LLMs being incredibly well trained on text, they often take the path of least resistance and may make their decision solely on the text based transcript and ignore numerical or other raw features from the other modalities. For example (P. Ekman and Friesen, 1978) would use the Action Units (AUs) approach when extracting features from user's facial expressions, the LLM may chose to ignore these features and instead focus on the text transcription.

To counter this, focusing on using DeepFace for the facial emotional recognition in which it would straight output an estimated emotion classification with confidence scores. I would then take the "dominant" emotion and confidence scores and pass this data to the LLM for the final classification. This in turn forces the LLM to process both the textual transcript and also the facial data in the favoured format.

2.16 LLMs and Robotic Conversations

Historically, emotion systems acted like basic matching tools, they would take the input then classify using a label such as [Happy] or [Angry]. Using LLMs allows for "Emotion Reasoning". Instead of just labelling the emotion, the LLM treats it like a reading comprehension task. This allows it to figure out the more complicated, mixed emotions for example, someone saying "I'm fine" whilst clearly showing distress and upset with their

facial expressions, then be able to reason through why they may be saying that and why their facial expression shows otherwise, and then offer a helpful response simultaneously. However, to keep the LLM from hallucinating or relying purely on text you have to anchor it with structured non-verbal data (Chaubey et al., 2026).

This is the kind of thing that bridges the gap between transitioning machines into basic reactive systems and now into empathetic companions able to support and offer advice to its users (K. Wang et al., 2020). The major challenge when running this live on a real robot is the delay conversationally in the interaction. Performing heavy data collection and then classification and analysis whilst also trying to keep the immersion of the conversational interaction between the user and the robot can prove very difficult, this will be discussed more in depth in a later section.

2.17 Emotionally Responsive Robots (Existing Systems)

When developing a social robot that is able to dynamically adapt to human emotions, there are some incredibly valuable insights when looking into the existing commercial systems, and seeing how they handle or even fail to handle these real time social interactions.

2.18 Traditional Platforms: Kismet and Pepper

To understand how far the field has come, research into the earliest pioneers of affective robotics must occur. Kismet (Breazeal and Scassellati, 1999), built by MIT in the late 1990s. Kismet at the time was a huge break through in the robotics and computing world, it proved that a robot doesn't need large complex language skills to portray empathy. Instead Kismet was able to do this with just basic facial movements (like moving its expressive eyebrows and ears) whilst also changing its vocal tone to match the human's mood. Whilst Kismet was advanced for its time, its system was driven by strict, pre-programmed rules rather than any actual artificial intelligence (AI), this meant it was never actually capable of holding any real conversations.

Decades later however, SoftBank Robotics introduced the world to Pepper (Pandey and Gelin, 2018), a mobile Humanoid Robot meant for real-world businesses. Pepper had on board the necessary hardware for capturing the multi-sensory data to understand a person's emotions. However, Pepper used a rule based system instead of AI or LLMs and so the conversation flow could get rather repetitive. Not only this, but Pepper also struggle with Latency issues. Pepper's replies relied heavily on slow, cloud based data processing. This would cause noticeable delays. In everyday conversations this delay of even a few seconds is often enough to break the immersion between the user and the robot.

2.19 The LLM Wave: Furhat and Abi

Stepping forward, more recently in the robotic world we have had more and more socially intelligent robots emerging. This is likely due to the more recent emergence of LLMs and AI. Furhat (Al Moubayed et al., 2012), developed by Furhat Robotics, unlike Pepper and the QT Robot (LuxAI S.A., 2026) Furhat has a unique "back-projection" system where a digital, fully animated face is projected onto the 3D physical mask. This allows the robot to display incredibly subtle, lifelike facial expressions and mimic human movements seamlessly. Because of this, researchers frequently use Furhat to test how people respond to expressive social cues. However, while Furhat provides an exceptional physical interface for displaying expressions, standard setups still require external AI logic to translate the raw, real-time camera inputs into actual conversational reasoning.

Abi, made by Andromeda Robotics (Brown, 2026), is amongst the newest generation of social humanoid robots. Abi solves the old school hard coded, rule based decision engine by replacing them with LLMs. Abi has currently been deployed to help combat loneliness, specifically within care homes for the elderly.

Abi uses machine vision to identify people along side a customised GPT-4 model to act as the engine. GPT-4 allows Abi to chat in a relaxed and natural manner. Due to the LLM running as the brain, Abi can dynamically adjust its entire personality based on the user it is talking to. For example, the Andromeda team build a compassionate "Dementia Mode" where the LLM alters the robot's vocabulary, pacing, and emotional cues to be more understanding, patient and supportive, allowing for the residents to talk to Abi like a real world friend.

2.20 Summary and System Motivation

The current state of these new social robotic companions are shifting away from the simple rule based classification approaches towards a new dynamic, open-ended emotion reasoning approach. As discussed previously, relying on a unimodal approach where the robot is only able to rely on the one sensory input, creates significant blind spots. The newer multimodal fusion is universally recognised now as the gold standard due to combining the multi sensory streams which therefore mirrors human perception and makes the emotional tracking of users more reliable.

2.21 The Gaps in Current Systems

While these new advanced humanoids use massive, complex engineering stacks in order to handle these multimodal streams, implementing an interactive emotional classification stack in a real-time setting introduces two well-documented practical challenges:

- **The Reality of Text-Bias (Modality Laziness):** In multimodal fusion literature, models often display a bias towards textual data. Due to Large Language Models possessing powerful linguistic pre-training, they naturally chose to take the path of least resistance. This means when it comes to using raw input data collected directly from the camera or using other methods such as Ekman’s AUs approach (P. Ekman and Friesen, 1978). To counter this, the focus shifted onto using DeepFace which outputs the user’s emotions as a level of confidence and then also a dominant emotion.
- **The Constraint of System Latency:** Running models locally can put a lot of strain on the hardware, especially if there are multiple different models running at once (LLM and DeepFace). This can increase the latency between the input received through the various input streams, and the output from the robot back to the user. In development this was a key concern as even a couple of seconds delay in response from the robot can break the natural flow of conversations and in turn the immersion.

2.22 Motivation for the Proposed Architecture

Rather than attempting to build a flawless, heavy-duty commercially ready solution. The focus became on these two practical constraints. To work within the limits of a compact, edge-deployed platform like the LuxAI QTrobot, the system would require a computationally efficient pipeline.

My simple engineering workaround addresses both constraints at once:

1. **Bypassing Latency:** Turning continuous video streams into low-dimensional numerical facial metrics completely avoids delay and complex reasoning the model requires to process raw visual tokens. This allows for these smaller models to run efficiently and still effectively.
2. **Balancing Modality Attention:** The model is explicitly prompted to output an emotional analysis conflict using (MISMATCH:), this forces the model to evaluate non-verbal numbers rather than cheating and following the lazier textual transcript approach as discussed previously. On top of this more intelligent models may even be able to detect the mismatch and thus interpret sarcasm, or other complicated emotions where facial expressions don’t always match the intended transcript.
3. **Unified Control Loop:** The model treats speech generation and robot behaviour control as a single task. In the speech output the model can also use inline command tags like so, gesture:NAME, expression:NAME. These allow the QTrobot to perform gesture and different expressions related to how they are responding.

Ultimately, this project provides a benchmark comparing a suite of open-source edge models, and evaluate how these small systems can handle classification accuracy and control

loops under these real world constraints. This research will also aid my in deciding the more suitable model for this kind of future work in social robotic interaction. This work also demonstrates a practical evaluation of how a lightweight, accessible architecture can be used to create a context-aware, empathetic social robot using a standard research robot.

Chapter 3

Design

This section will outline the design choices made in the emotion aware robotic system. I will cover the overall architecture, individual processing pipelines and the mechanisms in which the system utilises the use of LLMs to control the robot's output and its physical behaviour.

3.1 Architecture Notation and Conventions

The diagrams in this section follow a consistent set of visual conventions. Solid directional arrows represent synchronous, blocking calls, where one component waits for another to complete before then continuing. The dashed arrows indicates asynchronous communication, where data is passed between components without the sender needing a response. This was essential for the design of the system as the number one priority was to reduce latency to as little as possible. Separating these processes and having them working independently allowed for this.

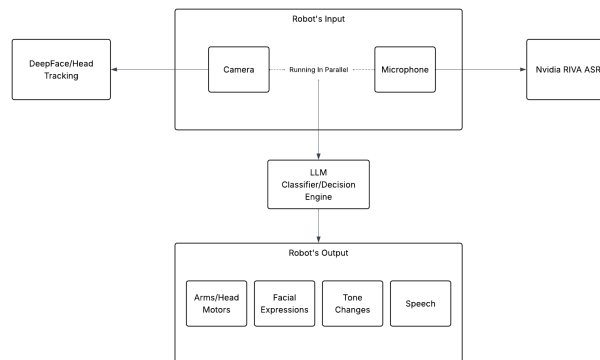


Figure 3.1: Full System Architecture

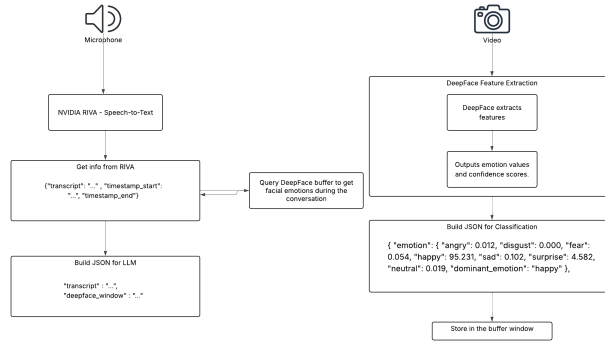


Figure 3.2: Audio and Video processing pipelines

3.2 High level System Architecture and Multimodal Fusion

At the highest level, the system follows a Sense-Think-Act (Siegel, 2003) structure. The robot senses its environment through a camera and microphone, it then processes that information to form a decision, and then acts through speech, movement, and facial expressions. The design challenge was ensuring that this all this could occur quickly enough so that the conversation would still feel natural, but also trying to avoid sacrificing any of the emotional depth that comes with that. To achieve this, running the audio and video processing in two independent pipelines operating in parallel would allow each process to run independently with neither of the process blocking one another. This can be seen in Figure 3.2.

The video pipeline uses DeepFace, a Python based facial analysis library introduced previously in the background, to continuously analyse frames from the robot’s camera. Each frame is broken down, feature extracted and then processes to output a set of emotion confidence scores across seven different categories (Ekman’s emotions + neutral). Rather than passing raw video frames into the LLM (which as previously mentioned would have caused issues regarding the way that LLMs are trained), these scores are then stored as a light weight numerical summary in a timestamped circular buffer. The buffer maintains a rolling window of the last 600 seconds of emotion data, continuously overwriting older entries as new ones arrive. This approach keeps memory usage constant regardless of how long the interactions last.

The audio pipeline on the other hand uses Nvidia Riva, a GPU-accelerated speech recognition service that runs locally on the robot’s Jetson AGX Orin hardware. Riva transcribes the user’s speech in real time. A new layer was then implemented to allow tracking via timestamps for when the user began and finished talking to the robot. These timestamps are the trigger points for the fusion step.

When Riva detects that the user has finished speaking, the system then uses those

timestamps to query the circular buffer. The window is then increased slightly by +2 seconds on either side to ensure that facial emotional data captured before the conversation are also used in the classification stage. This padding allows for the gathering of data that occurs just before the verbal content. This resulting emotional summary is then combined with the speech transcription to form a single structured input for the language model. This approach, fusing modalities at the point of decision rather than earlier in the pipeline, is broadly consistent with what the literature refers to as late fusion (Baltrušaitis, Ahuja, and Morency, 2018), and was chosen here to primarily avoid the need to pass visual data directly into the model’s context.

3.3 Node Topology and Software Orchestration

At the software level, the system is structured in such a way that a set of processes that communicate via the Robot Operating System (Quigley et al., 2009) middleware allows different parts of the system to run as independent nodes, publishing and subscribing to data channels without being directly coupled to one another. The camera feed, for example, is published on a ROS topic that both DeepFace facial recognition and head tracking can consume independently.

The conversational logic, receiving the transcript from Riva, then querying the facial buffer, constructing the prompt, and then passing this data to the LLM, is handled directly within a single orchestrator class. The language model itself runs on Ollama, a framework for running Large Language Models on consumer hardware. Ollama is used here rather than a cloud based API primarily for three reasons: it eliminates network latency, removes the reliance on a network connection to run, and doesn’t cost any additional funds.

3.4 LLM Control and Action Translation

By default, it is well known that LLMs generate outputs in natural language text. This is ideal when using it as a chat bot, but when it comes to controlling a robot and how the robot acts, changes vocal tone, expressions etc. Therefore, a mechanism is required to force the LLM to output explicit, machine readable commands. Bridging these two representations requires a translation layer that will convert this natural language into these commands.

The approach taken was to contain what the language model is allowed to produce. The prompt given to the model contained strict instructions on the fixed vocabulary of the supported actions. These actions contained gestures, facial expressions, and speech tones that the robot’s hardware was able to execute. The model is instructed to embed these commands directly in the output transcripts, placing the inline markup tags in the parts of the transcript where they were necessary. For example, the response might be structured like: “[expression:smile][tone:upbeat] That sounds really cool! [tone:calm] please tell me more.”.

The parsing layer would then process the model's output converting into the necessary actions for the robot. This markup-based approach was chosen over alternative approaches such as JSON output due to it allowing physical actions and speech to be interleaved naturally within the same response string. This make it straightforward to synchronise gesture timing with the speech delivery.

The system prompt also instruct the model to carry out explicit consistency checks before producing the response. The model is asked to evaluate whether the emotional content suggested by the facial data aligns with the sentiment expressed in the user's words. For example, identifying cases where a user's transcription appear positive in sentiment but their facial expressions suggest distress. The result of this check is included as a dedicated field in the model's output where the model is prompted to note that there is a mismatch and why. This is done so that the robot's response is more sensitive to cases where a user's verbal and non-verbal signals do not match, rather than solely relying on the transcription.

3.5 Interaction Flow and User Interface Design

The QTrobot platform in this project has several physical output channels available: An LED digital display that serves as its face, a motorised neck and arm joints, and a speaker. The interaction design aims to make use of all these outputs. All these channels are able to output simultaneously to produce a response that feels coordinated and natural rather than mechanical.

When the system begins a session, the robot enters a "watching and listening state", where the robot's head will track the user's head aiming to always be looking at the face of its user. During this state, the robot's microphone is active, listening for any speech from the user, whilst the camera is running DeepFace to gather expression and emotional data of the user. When the user begins speaking this is then listened and transcribed by Nvidia Riva, this then transitions the robot into a "processing" state where a transcript is created, then fused with the facial data, and the passed towards the language model. The model will process the SYSTEM_PROMPT and then classify the user's emotional state and provide a way in which the robot should respond to the user.

Once the response is parsed, the robot enters the "response" state and delivers it through those three simultaneous channels. The text-to-speech system vocalised the plain text whilst using the tone parameter to change the appearance of empathy from the robot. For instance, if the user's detected as being down or feeling sad then the the robot should respond speaking more slowly and softly. At the same time, any expression tags in the response are rendered on the LED screen on the head of the robot, thus changing its facial expression, and finally any gesture tags trigger the corresponding motor commands. Once this is complete the robot then returns to the "watching and listening" state where it is ready and waiting for its next input.

This multi-channel output was a deliberate design choice rather than a technical

requirement. A text only response would have been easier to implement, however using the robot's physical expressiveness alongside its speech is more consistent with natural emotional communication between people, and thus deemed important for the social presence and improving the immersion.

Chapter 4

Implementation

This section will describe the practical implementation of the robotic interaction system. The system was implemented as a combination of existing ROS-based robot nodes and custom Python orchestration code. The ROS nodes allowed access to the robot’s main sensing components, including the camera stream, Nvidia Riva speech recognition, and DeepFace facial emotion recognition. The custom Python implementation was responsible for managing the data being collected from these nodes, processing said data, synchronising, then constructing the input with the data for the LLM, and then converting the response into the robots’ actions.

The implementation followed the multimodal structure described in the system design. Audio and video processing were kept separate so that speech recognition and facial analysis could run in parallel to each other. The video pipeline continuously analysed the user’s facial expressions and then stored these results in the timestamped ring buffer. The audio pipeline transcribed the user’s speech, recording the start time and end time of each utterance. Once the user had finished speaking these timestamps would then be used to query the buffer to retrieve the matching facial emotion data from the buffer. This data was then summarised and combined with the transcript before then being passed to the language model for classification and response.

4.1 Runtime Architecture

At runtime, the system consists of several independent components. The low-level sensing and recognition functionality, which is handled by ROS nodes, while the higher-level decision logic was implemented in Python. This allows the main orchestration code to use the output from the existing robot node without directly coupling all components into a single program.

The camera, DeepFace and Nvidia Riva components were run as separate processes. DeepFace continuously processed the camera stream and produced facial emotion results. Nvidia Riva processed microphone input and produced speech to text transcriptions. The

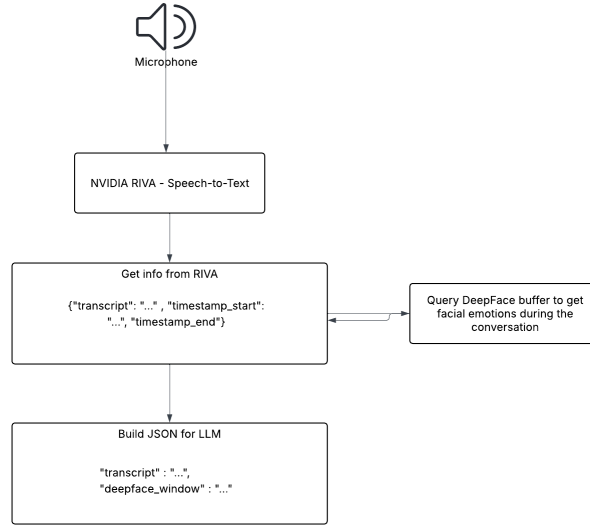


Figure 4.1: Audio Processing Pipeline

custom Python Script then used these outputs to build the multimodal input required by the LLM.

This structure allowed audio and video to be processed concurrently. DeepFace analysis could then occur while the robot was listening and even responding to the user. Similarly, the speech recognition process did not need to wait for the video processing to complete. This was important to ensure the system maintained a natural flow of conversation reducing the delay as much as possible.

4.2 Audio Processing

The audio processing pipeline used Nvidia Riva for automatic speech recognition and transcription. Riva received the microphone input from the robot and then returns the transcription results. A Python layer was implemented around the Riva output to extract the final transcript and recording the timing of each user’s utterance.

For each completed utterance, the system stored three key pieces of information: the transcript, the timestamp marking when the user began speaking, and the timestamp marking when the utterance came to an end. These timestamps, as previously mentioned, are then used for gathering the facial emotion data from the ring buffer during that time period. See Figure 4.1

The result audio object contains the spoken text and its associated time window. This allows the system to treat utterance as a discrete interaction unit. Once a final transcript was available, the audio pipeline triggered the fusion stage, where the DeepFace data,

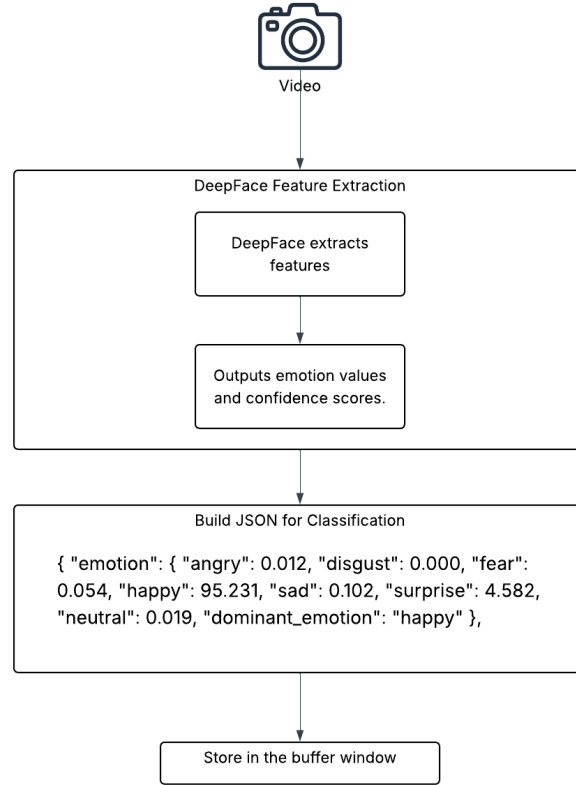


Figure 4.2: Video Processing Pipeline

corresponding to that time window, was then retrieved from the emotion buffer.

4.3 Video Processing

The video processing pipeline relies on DeepFace to analyse the user’s facial expressions. DeepFace is configured to run at 5 frames per second using the VGG-Face model. This frame rate was chosen so that the robot was still able to provide regular emotion updates whilst still avoiding the processing cost of analysing every single frame from the camera feed.

Each DeepFace result contained the detected dominant emotion (the emotion with the highest confidence scores) and then a set of all the confidences scores for all the other emotion categories (neutral, happiness, sadness, fear, disgust, anger, and surprise). Each result was stored with the timestamp so that it could be letter retrieved when the user began speaking. The system stored the numerical emotion output rather than the raw video frames. This ensured that not only less memory was used up, but also the data was easier to pass to the language model. Each stored result therefore represented a lightweight summary of the user’s

facial expression at a specific moment in time.

4.4 Emotion Buffer

The DeepFace results were stored in a timestamped buffer with a maximum window of 600 seconds. this buffer acted as a rolling window of recent facial emotional data. As new DeepFace results were added, older entries would be discarded once they exceeded the 600 seconds window. This ensure that the robot was able to keep a good chunk of facial data that could be used when required but also ensured that not too much memory was then taken up.

Since DeepFace was configured to run at 5 frames per second, this means the buffer would end up storing approximately 3,000 entries when full. this remained manageable as each entry only contained a timestamp, dominant emotion label, and numerical emotion scores. For this reason, no image frames were stored in the buffer.

The purpose of this buffer was to allow the system to retrieve facial emotion data from the same time period of the spoken utterance. this avoided the need for the audio and video pipelines to be directly synchronised at every moment. Instead, both pipelines produced timestamp outputs, meaning the fusion stage could easily pull the relevant data based on the timestamps to then build the input for the language model.

4.5 Multimodal Fusion

Multimodal fusion occurred when Nvidia Riva returned a final transcript. the start and end timestamps for the utterance are used to query the DeepFace buffer. This query retrieves all facial emotion entries that occurred during the time window that the user was speaking through.

The selected DeepFace entries where then analysed before being passed to the language model. This was done to reduce the amount of data sent the LLM and provide a clearer summary of the user's facial expression during the utterance.

The emotion window was summarised in three ways. First the system counted the number of times each emotion appeared as the dominant emotion. This gave an indication as to which emotion was most frequently detected while the user was speaking. Secondly, the system would then calculate the average score for each emotion across the selected window. this represented the general emotional trend during the utterance. Lastly, the system then recorded the peak score of each emotion. This meant short but strong emotional expressions could still be preserved in the summary.

The final emotion summary contained the dominant emotion counts, average emotion scores, and peak emotion scores. This summary was then combined along with the transcript and used as the input for the language model.

4.6 Language Model Integration

The language model component of the robot was run locally using Ollama. Ollama utilises the Jetson hardware running natively in the robot to decrease latency for LLM response that may have occurred over the internet. Ollama allows different models to be pinned and test against each other without needing to change code or any of the pipeline. The comparison of these models will be detailed in the testing and evaluation section of this report.

The LLM was used as the main decision engine/classifier for the system. It would receive the user's transcript, summarised facial data, and a system prompt describing the model's role in the system and how it can control the robot. The prompt instructed the model to consider both the verbal content and non verbal emotional data when generating a response.

The model input was structured so that the transcript represented what the user said, whilst the DeepFace summary represented how the user appeared during the interaction. This allowed the LLM to identify the environment and also situations where facial emotion data conflicted with the user transcript. For example, if the user said they were fine but the facial emotion summary showed high sadness or fear scores, then the model would be able to take this into account and provide a more sensitive response.

4.7 Prompt Construction

The prompt sent to the LLM contained the system instruction, the user transcript, the facial emotion summary, and the list of supported robot actions. The system instruction defined the behaviour expected from the robot and restricted the model to a fixed set of output commands. How successful the model was in sticking to the prompt and instructions was also evaluated and will be discussed in the testing and evaluation section also.

The prompt also instructed the model to return its response using the inline action markup (for example, [expression:sad]). These tags allowed the model to specify the robot's behaviours alongside the spoken response. this made it possible for the robot's speech and physical behaviour to be dynamic and coordinated as part of the same generated output.

4.8 Action Parsing

The LLM output contains a combination of natural language and inline robot action tags. A parser was developed to separate the parts of the response and then perform the required actions. Natural language text is sent to the speech system, while the action tags are converted into robot commands.

The action tags represented the robots outputs that the LLM is given access to control. They include: facial expressions, arm and head gestures, speech, and tone. For example, the

model could generate response that instructed the robot to smile, speak in an uplifting tone and perform arm and head gestures whilst delivering the spoken response.

The parser acted as a translation layer between the LLM and the robot hardware. This is necessary because the LLM output could not be sent directly to the robot. Each generated tag has to be identified, checked if it is a valid command, and then mapped to the supported command.

4.9 Robot Output Control

The final stage of the implementation converted the parsed response into the robot output. The robot could respond using any of its facial display images, arm movements, head movements, speech system and tone control.

Facial expression tags are mapped to the robot's display based expressions. Gesture tags were mapped to predefined movements on the robot, these include movements such as head gestures (e.g. shaking the head) or simple waving hand motions. Speech text was sent to the robot's text-to-speech system, and tone tags were used to change the vocal delivery of the robot.

This allowed the robot to produce a response that was not limited to simple spoken text. instead, the robot could combine speech, facial expression, head movement, and tone to create a more expressive and potentially natural interaction. The use of inline markup made it possible for these actions to be coordinated and linked to specific parts of the response rather than being performed separately to the speech.

4.10 Implementation Challenges

One of the main implementation challenges was latency. The system needed to perform the speech recognition, facial emotion analysis, emotion summarisation, LLM inference, response parsing, and robot output control quickly enough to not break the natural interaction. This was addressed by keeping the audio and video pipelines independent of each other and allowing them to run in parallel.

Another challenge was aligning the audio and video data. the system needed to use facial emotion data that corresponded to the same period as the user's speech. This was addressed by timestamping both Riva speech transcription and DeepFace outputs. Then using these timestamps to query the DeepFace buffer.

A further challenge was deciding how much facial emotion data should be passed to the LLM. Passing every single frame that DeepFace produced would have increased the prompt size, increasing latency, and increasing noise for the LLM. This could have caused the LLM to go down that route of least resistance that was previously talked about, causing it to miss mismatching data. To avoid this the emotion window was summarised using the

dominant emotion counts, average emotion scores, and peak emotion scores. The system also needed to be able to handle the unpredictability of the LLM output. Although the prompt restricted how the model could respond and act with the robot, the model could still produce malformed or unsupported tags. The parser therefore validated each tag before performing the action.

Finally, the facial emotion recognition has its practical limitations. DeepFace output can be affected by lighting conditions, camera angle, occlusion, and whether the user is facing the robot or not. For this reason, the system treated these facial emotion analysis as contextual information rather than a definitive measurement of the user's emotional state.

Chapter 5

System In Operation

This section describes how this complete system operates during a live interaction. The focus will be on the sequence of events that occur when a user interacts with the robot.

The system operates as a continuous interaction loop. Initially, the robot enters a watching and listening state. In this state, the camera and microphone, and speech recognition and facial emotion recognition pipelines are all active. The robot gathers data on the user's face using DeepFace and stores it in the buffer. During this process and throughout this phase the robot will attempt to maintain eye contact with the user to help with the interaction whilst also allowing the robot to keep gathering facial data.

When the user begins speaking, Nvidia Riva processes the microphone input and produces the transcript. The Python script then logs the timestamps of this utterance. At the end of the user's speech input, the system then queries the DeepFace buffer to get the emotional state across that interaction. Once this data is pulled from the buffer it is summarised and combined with the transcription, ready to be passed to the language model.

A system prompt along with the interaction data is passed to the large language model. The large language model then uses this prompt and information to classify the user's emotional state, and then generate an appropriate response. This response is then broken down and parsed.

Before the response can be executed the parser separates the action tags from the natural language text. Each action tag is checked and validated to ensure that the robot is able to perform the action or expression. This ensures the robot doesn't run into any errors that the LLM may cause if generating incorrect tags.

The robot then enters the response state. During this state, the text-to-speech system vocalises the response, tone commands adjust the delivery style, facial expression commands change the robot's display and gesture commands trigger movements with the head and or arms. These outputs occur as part of the same response sequence generated by the LLM in the order they need to be executed. The commands are split into blocking and none-blocking commands and gestures. This means some of the commands such as "shaking the head" or

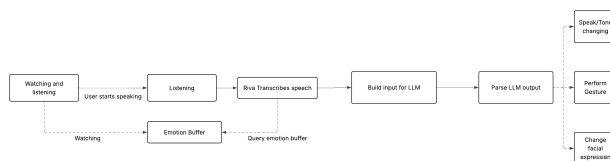


Figure 5.1: State Diagram

”winking” doesn’t break immersion, whilst other commands such as ”waving” or ”scratching head” can still occur whilst the speech continues keeping the natural flow of the interaction.

After the response is finished, the robot returns to the watching and listening state. The camera and microphone remain active, and the system waits again for the user’s next interaction. This loops around and continues through the previously described states ensuring to append the conversation to the “conversation_history” variable so that the user can continue to reference previously mentioned topics in the conversation. This concludes the repeated interaction loop where each user input is processed, fused with the facial emotion data, passed to the LLM, and then converted into multimodal robot behaviour. See Figure 5.1 for the state diagram.

Chapter 6

Testing and Evaluation

This section focuses on systematically evaluating and benchmarking the performance of various open-source Language Models within the software pipeline of the QT social robot. To ensure a robust assessment of each model's viability for real-time, empathetic human robot interaction (HRI). In testing, I moved beyond a basic accuracy test and instead tested three critical things:

- **Modality Integration (Unimodal vs Multimodal):** An in depth analysis, assessing whether the integration of DeepFace for the facial emotion recognition improves the the classification results.
- **Core Classification Performance:** How capable are the different models at correctly classifying these emotions? In this test we use the system to classify the data in the MELD dataset across all the models.
- **Operational Robustness and Instruction Following:** Here, a series of tests are run on the LLMs to understand which models are best for our robot. These test range from: robustness, instruction compliance, length of response, and more.

In this section the aim is to vigorously test these models to find the best for our use case. The aim is to find a model that is able to be fast enough for real conversational flow, reliable enough to follow the prompt accurately, and smart enough to classify the user's emotional state and provide a suitable response.

In all test where we are using DeepFace for the facial emotion recognition we used the MELD data set (Poria et al., 2019). This is a dataset that consists of lots of clips from the TV show "friends". To evaluate the system, a subset of 100 clips from MELD was utilised. MELD allows for rigorous stress testing. While this presents a high unconstrained environment compared to a standard, front-facing human-robot interaction, it serves as an effective baseline to benchmark the model's ability to process conflicting or noisy multimodal inputs.

6.1 Unimodal vs Multimodal Emotion Recognition

A core objective of this study was to evaluate and understand how fusing multiple sensory inputs impacted the emotion classification of various language models. Specifically, we compared a text-only (unimodal) approach against a multimodal setup that combined the text transcripts with facial emotional data.

To analyse these models effectively splitting into two testing phases was the way forward: an initial baseline test where the prompt told the model to treat both the textual and vision data equally. The second test then used a stricter prompt that prioritised text more highly over vision due to noise created by incorrect visual data classifications.

6.1.1 Test 1: Baseline Evaluation and the "Multimodal Penalty"

In the first test, the system instructed the models to treat the text transcript and the facial tracking data equally. In this setup results indicated that adding the visual data caused a huge drop in accuracy across all of the seven models tested. See Figure 6.1

Table 6.1: Test 1: Unimodal vs Multimodal Accuracy (%)

Model	Unimodal Accuracy (%)	Multimodal Accuracy (%)	Accuracy Delta (%)	DeepFace Baseline (%)
mistral_7b	58.0	29.0	-29.0	11.0
gemma3_4b	42.0	9.0	-33.0	11.0
granite3_2_2b	40.0	11.0	-29.0	11.0
llama3_2_3b	33.0	16.0	-17.0	11.0
qwen2_5_3b	19.0	7.0	-12.0	11.0
phi4-mini_latest	16.0	2.0	-14.0	11.0
smollm2_latest	4.0	2.0	-2.0	11.0

As you can see in Figure 6.1 the DeepFace model achieved a baseline accuracy of 11% on the MELD dataset which demonstrates how poorly the DeepFace model performed independently. This poor baseline questions the ecological validity of the MELD dataset. MELD consisting of television clips featuring frequent camera cut off, profile angles and having multiple subjects in frame. In contrast intended deployment of the QTrobot will be in the an environment where the single user is positioned directly in front of the robot's camera in a fairly controlled setting. Therefore, the poor visual feature extraction observed here is likely amplified by the dynamic nature of the dataset. The resulting "Multimodal Penalty", where models struggled to reconcile accurate text with inaccurate visual data, highlights a critical vulnerability in smaller LLMs, however the raw 11% baseline should not be viewed as the expected performance ceiling for live, face to face robotic deployment.

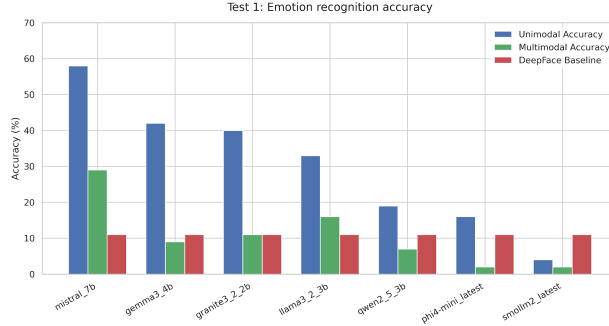


Figure 6.1: First test demonstrating Emotion Recognition Accuracy across the seven models

When this low-quality numerical data was fed into the language models it took some of the attention away from the textual input from the transcription. Instead of using the visual data as a helpful back or as supporting evidence to the transcription the model trusted the flawed data over the highly accurate transcripts.

As you can see in Figure 6.1 this resulted in severe performance drops:

- Mistral 7B fell from a leading unimodal accuracy of 58% to just 29% in the multimodal setup.
- Gemma3 4B suffered a 33% collapse, dropping from 42% to 9%.
- Granite3.2 2B fell from 40% to 11%, effectively matching the poor 11% baseline of DeepFace.

This failure is highlighted by the Multimodal Mismatch Rate. For almost every model under 7B parameters, the mismatch rate sat between 95% and 100%. This shows that when models faced conflicting data between the text and the face tracking, they consistently chose the wrong facial data over the corrected text data.

6.1.2 Test 2: Prompt-Engineered Mitigation

To fix the previous attention hijacking issue found in Test 1, I ran a second experiment with a new prompt. The new prompt contained much of the information from the original prompt but then an added section about the DeepFace data it would receive. It specifically notified the model that the primary source of truth was the textual transcription and that the visual data could be prone to errors and adding in extra noise. The DeepFace data was only to be used as a tie breaker or solidifying decisions being made on the transcription data.

The results showed a clear divide based on model size:

1. High-Parameter Models: Larger models successfully followed these instructions and filtered out the visual noise. Mistral 7B reduced its multimodal penalty from -29% to

Table 6.2: Test 2: Unimodal vs Multimodal Accuracy (%)

Model	Unimodal Accuracy (%)	Multimodal Accuracy (%)	Accuracy Delta (%)	DeepFace Baseline (%)
mistral_7b	55.0	41.0	-14.0	11.0
granite3.2_2b	50.0	28.0	-22.0	11.0
gemma3_4b	44.0	9.0	-35.0	11.0
llama3_2_3b	36.0	13.0	-23.0	11.0
phi4-mini_latest	32.0	3.0	-29.0	11.0
qwen2_5_3b	27.0	8.0	-19.0	11.0
smollm2_latest	3.0	2.0	-1.0	11.0

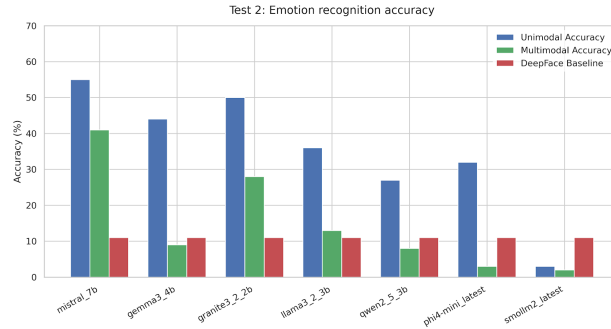


Figure 6.2: The second test outlining the accuracy of the seven models with the updated software.

-14%, bringing it's accuracy back up to 41%. Its mismatch rate also dropped by 39%, proving it actively ignored the bad visual data. Granite3.2 2B showed similar improvements dropping it's mismatch rate to 71% and recovering its accuracy to 28%.

2. Small Language Models: On the other hand, models under the 4B parameters (like Gemma3 4B, Qwen2.5 3B, and Phi4-Mini), excluding Granite3.2 2B, showed almost no recovery. their mismatch rates stayed stuck between 89% and 100%, and their accuracy didn't break past 13%. Even with strict instructions to ignore the noise data these smaller models simply lacked the reasoning capability to disregard structured numerical inputs when they conflicted with the textual transcriptions.

6.1.3 Robotic Deployment Implications: Latency vs Compliance

For deployment on a live social robot like QT it is essential to balance the accuracy against system stability and then also conversational speed.

- **Format Compliance:** Giving the models complex multimodal data and long

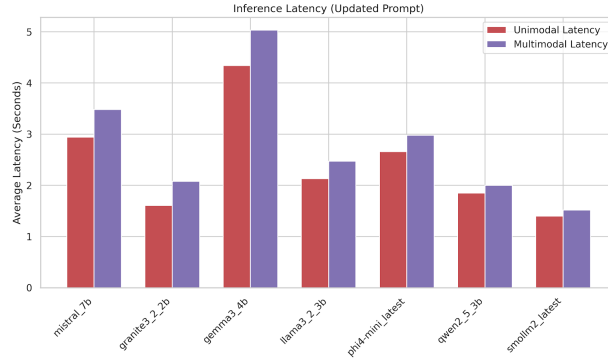


Figure 6.3: Chart demonstrating the Latency differences between the unimodal approach and Multimodal approach.

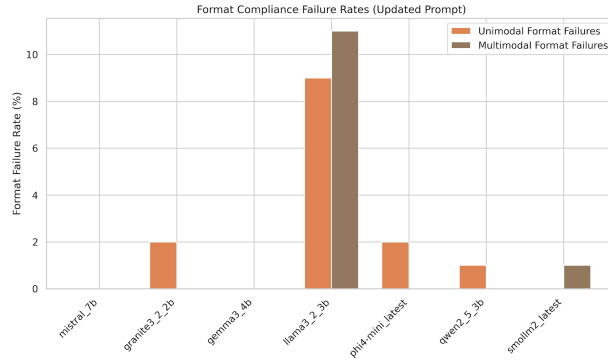


Figure 6.4: Chart demonstrating the Format Compliance failure rates.

instructions did not break the software pipeline. The best models (Mistral 7B and granite3.2 2B) maintained a 0% failure rate. This ensured that inline robotic commands remained readable by the system and allowing the robot to execute the gestures and emotions.

- **Latency Trade-off:** While Minstral 7B was the most accurate model, it was slow in comparison to some of the other models (excluding Gemma3 4B). Granite3.2 2B on the other hand emerged as a much better candidate for real-world conversational flow due to its processing in just 2.08 seconds whilst still being able to successfully handle the noisy DeepFace data.

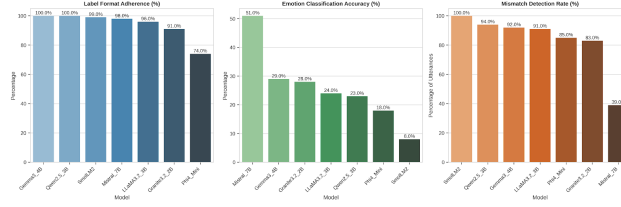


Figure 6.5: Comparative charts illustrating Label Format Adherence, Strict emotion Accuracy, and Mismatch Detection Rate across all models.

6.2 Multimodal Emotion Classification and Mismatch Detection Results

6.2.1 Baseline Data Distribution and Evaluation Metrics

In order to thoroughly test the reasoning capabilities of the models, I evaluated the 7 models using a subset of 100 clips from the MELD dataset. In this test focus was on three main metrics: label format adherence (how well the model stuck to the 7 allowed emotion categories), strict emotion classification accuracy, and the cross-modal mismatch detection rate (how often the model flagged a conflict between the text and facial features).

Before looking at the model scores, it is important to understand that the sample set is heavily imbalanced towards one category.

- Neutral: 52 percent • Surprise: 15 percent • Joy: 13 percent • Anger: 9 percent • Sadness: 8 percent • Fear: 2 percent • Disgust: 1 percent

This imbalance is important because looking only at raw accuracy can hide underlying biases in how the models guess. The overall performance metrics are below.

6.2.2 The Majority-Class Illusion

At first glance, Mistral 7B looks like the best model for reasoning and producing the desired output. It achieved 51% accuracy, beating the next best model by 22%. However, a deeper look shows the high score is actually the Majority-Class Illusion caused by guessing the majority class.

Mistral 7B labelled 65% of all inputs as "neutral". Due to the dataset itself being 52% neutral, Mistral played it safe and just guessed the most common option.

The confusion matrix in Figure 6.6 demonstrates that Mistral 7B basically acts a single-class predictor. It correctly guesses neutral but fails to recognise minority emotions like surprise, fear, or disgust. Because it treats almost everything as neutral it rarely flags conflicts between the transcription and the DeepFace data. This explains the artificially low 39% mismatch detection rate.

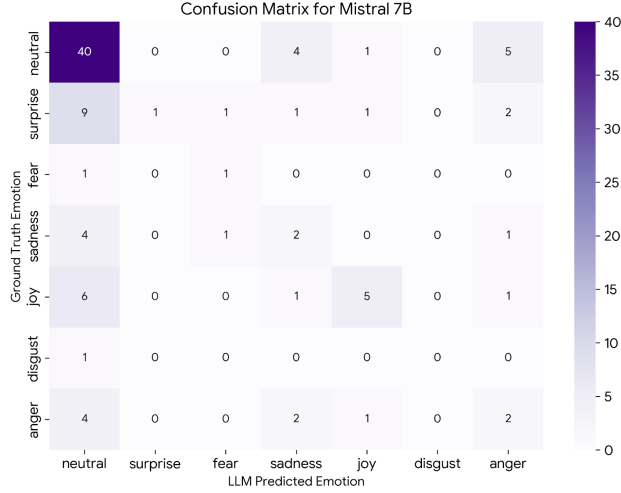


Figure 6.6: Mistral 7B Confusion matrix demonstrating a high concentration of neutral classifications alongside misclassification of minority emotional states.

6.2.3 Evaluating True accuracy via the Macro F1 score

To counter the previously mentioned imbalance in the dataset ("neutral" being 52% of the dataset) I used the Macro F1 score to evaluate the performance of the models. Unlike strict accuracy, the Macro F1 score allows us to evaluate the precision for each of the emotional classes. This evens out the weights for the minority emotions. In this case it severely penalises models that exploit the majority classes while ignoring the minority classes (in this case "fear" and "disgust").

The resulting Macro F1 scores:

- Mistral 7B: 30.10%
- Qwen2.5 3B: 27.61%
- Gemma3 4B: 25.16%
- Granite3.2 2B: 24.61%
- LLaMA3.2 3B: 18.78%
- Phi4-Mini: 9.93%
- SmolLM2: 2.16%

The use of the Macro F1 score shatters the illusion of Mistral 7Bs dominance. While it achieved a superficially high strict accuracy of 51%, its Macro F1 score plummets to 30.10%.

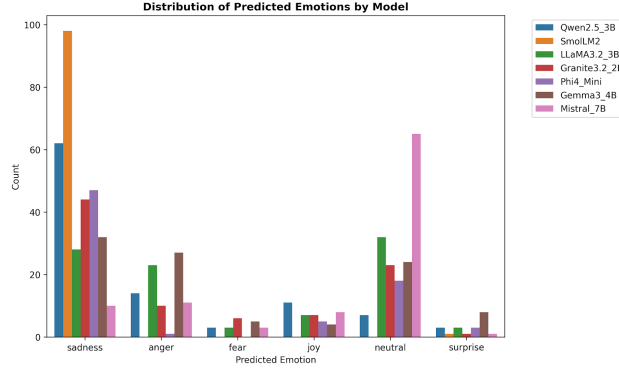


Figure 6.7: Showing the total distribution of predictions across all emotions. Highlighting "sadness" predictions from the smaller models.

This confirms the suspicions talked about previously. Mistral 7B functions largely as a majority-class predictor and fails to demonstrate a robust, balanced understanding across the human emotional spectrum.

Furthermore, models that suffered from the extreme behavioural collapse, specifically SmolLM2 (defaulting to 98% sadness), were heavily penalised, resulting in a near zero score of 2.16%.

Ultimately, none of the models managed to achieve a Macro F1 score exceeding 31%. This realistically demonstrates that these out-of-the-box Language models, without task-specific tuning or better emotional facial data, lack the discriminatory power to be able to reliably reason and offer successful emotional classifications for my Multimodal approach.

6.2.4 Behavioural Collapse and Sentiment Trapping in Small Language Models

In contrast, the smaller models failed in the exact opposite way. They collapsed into negative sentiment loops. SmolLM2 and Qwen2.5 3B are perfect examples of this.

Instead of guessing the majority class ("neutral") like Mistral 7B, the smaller models instead became hyper fixated on "sadness".

- SmolLM2 classified 98 percent of all samples as "sadness."
- Qwen2.5 3B classified 62 percent of all samples as "sadness."

Because SmolLM2 constantly decided to guess "sadness" this meant anytime the textual prompt or facial data was not sad, the model was flag a mismatch. This explains the 100% mismatch detection rate. This isn't a good sign of reasoning and seems to be a complete architectural failure. This model most definitely should not be used in the robot.

Model Identifier	Label Format Adherence (%)	Strict Emotion Accuracy (%)	Mismatch
Gemma3 4B	100.0%	29.0%	92.0%
Qwen2.5 3B	100.0%	23.0%	94.0%
SmolLM2	99.0%	8.0%	100.0%
Mistral 7B	98.0%	51.0%	39.0%
LLaMA3.2 3B	96.0%	24.0%	91.0%
Granite3.2 2B	91.0%	28.0%	83.0%
Phi4-Mini	74.0%	18.0%	85.0%

Table 6.3: Label Format Adherence and Strict Emotion Accuracy (%)

6.2.5 Cognitive Load and Syntactic Adherence

I also tested how well the models could stick to the given formatting rules whilst under cognitive strain. Models were to output exactly one label from the strict list of seven emotions. While Gemma3 4B and Qwen2.5 3B maintained a perfect 100% adherence rate, smaller models struggled with this significantly.

The worst collapse happened with Phi4-Mini, which dropped to a 74% adherence rate. Instead of outputting clean, single words, it broke the rules and generate long, messy descriptions like "mixed (neutral/sadness)" or "sadness or anger (based on facial expression)."

This shows that with the smaller models (often under the 4B parameters) trying to do too many things at once, such as reading text, processing visual numbers, checking for conflicts and then reasoning, causes them to become overloaded. To handle the cognitive load, they sacrifice their formatting, which in turn then breaks the system.

6.3 Initial LLM Benchmarking and Evaluation

To figure out the best "brain" for the QT robot, I benchmarked the seven LLMs against each other. The goal was to find out which models could handle the real-world Human-Robot interaction (HRI) requirements the best.

6.3.1 Format compliance and Structural Reliability

Some of the previously covered tests actually already did a short evaluation on this but the primary test in those sections were not for the format compliance.

For a robot to move, the LLM must output text in very specific inline markup commands. if the formatting is wrong then the script's parser may fail to find and execute the commands. During testing, Qwen2.5 3B, Mistral 7B, and Gemma3 4B achieved perfect 100% format compliance. However, Llama3.2 3B and Granite3.2 2B struggled, scoring only 80% and 70%. granite often completely dropped mandatory formatting lines, while Llama missed crucial

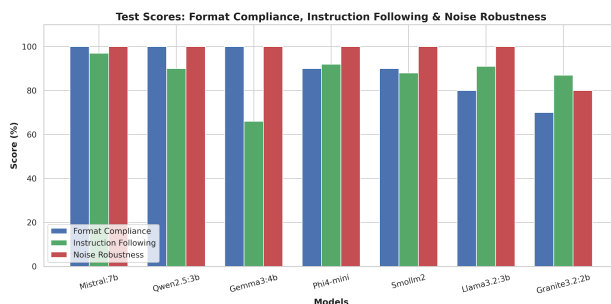


Figure 6.8: Performance Comparison of the models across format compliance, instruction following, and input noise robustness.

syntax. In a live robot a 20% to 30% failure rate is dangerously high, making these models not suitable for my use case.

6.3.2 6.3.2 Instruction Following and Noise Robustness

Next I tested how well these models mapped human emotions to physical robot gestures, and how they handled noisy inputs (such as filler words, slang, and typos).

Almost all models handled noise perfectly, scoring 100%, except for granite3.2 2B, which failed 20% of the time on slang heavy inputs. Instruction following, making decisions logically about which gestures to use, had more varied results. Mistral 7B was the most accurate at 96%, followed by Phi4-Mini (91%) and then Qwen2.5 3B (90%). Despite having perfect formatting, Gemma3 4B struggled with logic, scoring only 66% because it kept generating conflicting physical expressions.

6.3.3 Conversational Consistency

To make sure the robot's voice sounds natural, the models need to output consistent response lengths. Unpredictable lengths cause erratic conversation that can often swing from unnaturally short to insanely long.

Qwen2.5 3B, Gemma3 4B, and Phi4-Mini kept their responses very consistent. On the other hand, Granite3.2 2B and SmolLM2 were highly unpredictable, with Granite3.2 2B often generating outputs so long they triggered system errors 20% of the time.

6.3.4 Latency Processing

Latency (the delay) is the biggest hurdle in social robotics. For conversations to feel natural the robot needs to respond in under 2.0 seconds.

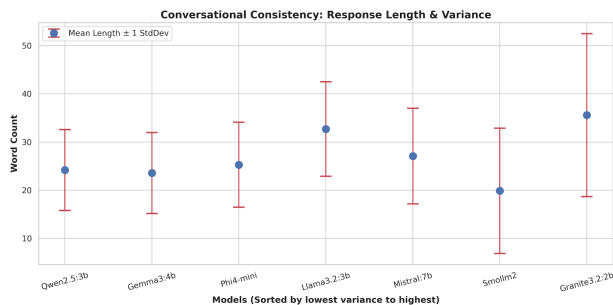


Figure 6.9: Error Bar Chart showing Conversational Consistency. Specifically the response length and Variance.

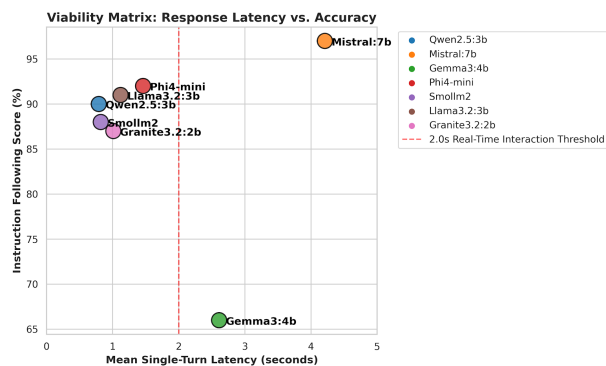


Figure 6.10: Plot describing the latency against instruction following accuracy. The dashed red line is the 2 second target keeping the conversation natural.

While Mistral 7B was the smartest model, it was far too slow for a real robot and to keep the natural social interaction, averaging at 4.21 seconds per response. Gemma3 4B was also slightly too slow at 2.61 seconds.

Qwen2.5 3B emerged as the victor here. It averaged a rapid 0.79 seconds per response and stayed incredibly stable even as conversation history increased easily beating the 2 second threshold.

6.3.5 Initial Conclusion

Based on this benchmarking, Qwen2.5 3B is the most robust model for our robotic system. It is the only model that guarantees perfect formatting for command outputs, it makes logical decisions, and processes text fast enough for the real-world conversations.

Chapter 7

Conclusions

This project aimed to develop a low latency, multimodal emotion recognition pipeline utilising the QTrobot platform, integrating it directly with Large Language models (LLMs). This implementation successfully demonstrated that decoupling audio and video processing, by introducing a timestamped rolling buffer for facial emotional data, effectively mitigates the latency typically associated with a multimodal fusion approach.

However, the extensive benchmarking across the seven local LLMs revealed a significant "Multimodal Penalty". When the models were required to reconcile these textual transcripts with numerical visual data, smaller models (sub-4B parameters) frequently suffered from a behavioural collapse, defaulting to negative sentiment loops. On the other hand, larger language models such as Mistral 7B masked its inability to fuse data through a majority class bias, predominantly defaulting to neutral classifications.

Despite these limitations across all tested models, Qwen2.5 3B emerged as the most suitable option for the system architecture. It was able to successfully balance short latency with strict formatting compliance. This ensures the reliable execution of physical robotic gestures and expressions during the real-time interaction.

7.1 Limitations

7.1.1 Hardware and Software Dependency Constraints

The physical deployment of the architecture onto the Jetson AGX Orin hardware highlighted some significant constraints. To maintain a structural stability in using the Robot Operating System (ROS) to control nodes and services across the robot, the local Python version had to be kept strictly to an older version (3.8). Specifically, attempts to then try and implement more sophisticated facial emotion recognition models such as Baltrusaitis et al. (2018) would fail due to the EfficientNet requirement crashing repeatedly at runtime. This was also due to compatibility issues with the Jetson hardware. Because of this the processing pipeline was

restricted to using DeepFace.

7.1.2 Dataset Viability and Selection Criteria

The reliance on the Multimodal Emotion Lines Dataset (Poria et al., 2019) was a compromise forced by deficiencies in alternative multimodal datasets. Initial evaluation of the CMU-MOSEI dataset’s official download link failed. A route around this found downloading the dataset from a third party was successful but in turn there were now ground truth labels for the data thus preventing a reliable validation of the system pipeline. Furthermore, the RAVDESS dataset relied on actors repeating identical, semantically neutral phrases across various facial emotional expressions. Due to the linguistic context remaining static this wouldn’t have evaluated the system as a whole but instead just DeepFace. This left MELD. While MELD provided the necessary cross-modal variance, the unconstrained television broadcast format introduced large amounts of background noise and camera cuts. This in turn degraded the performance of DeepFace. The original hope was that due to DeepFace’s built in analysis pipeline where it finds the face, crops the entire image down and analyses just the phase could counter this issue. As we can see by the resulting baseline of 11% it was not successful.

7.2 Future Work

7.2.1 Middleware Improvement and Advanced Computer Vision

To overcome some of the hardware and software barriers imposed by the legacy dependencies on the robot, a critical next step is migrating the entire system architecture to ROS 2 (Robot Operating System 2). This migration will allow for supporting more modern Python Distributions and in turn allow for newer and more robust technologies to be used with the robot.

With this updated software stack, future iterations of computer vision models could replace the VGG-Face model used in DeepFace, or even supplementing OpenFace 3.0 in instead. This will allow for more accurate results when it comes to the facial emotional data.

7.2.2 Incorporation of Acoustic Speech Emotion Recognition (SER)

To resolve the systematic blind spot and improve the multimodal pipeline of the system, the introduction of a local Speech Emotion Recognition (SER) model, such as Wav2Vec2 (Baevski et al., 2020) , would allow the system to extract raw audio features directly from the user’s speech. By passing vocal parameters such as pitch variance, speech intensity, and valance alongside the textual transcription, the language model will gain more context required to identify more complex emotional states such as sarcasm or hidden distress.

7.2.3 Transitioning to Live Human-Robot Interaction (HRI) Trials

While offline benchmarking against standard datasets was necessary for validating decisions and choices in development of the QTrobot, to evaluate true efficacy of an empathetic social robot, this requires live user studies. Future work in this aspect would involve deploying the system using the optimised Qwen2.5 3B pipeline in a real-world environment with real users. The trials would allow the gathering of data both quantitative and qualitative, using standard HRI evaluation techniques and frameworks.

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